SleepWise Expert System

Test Cases Document

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DCIT 313: Introduction to Artificial Intelligence | University of Ghana

189	78	267
Knowledge Base Tests `test_knowledge_base.py`	Integration Tests `test_system_integration.py`	Total Tests **All PASSED**

This document lists all test cases used to validate the SleepWise Knowledge-Based System. Tests are organised by script and section. Each row shows the test ID, description, input condition, expected result, actual result, and verdict. All 267 test cases passed.

PART 1 — Knowledge Base Tests (test_knowledge_base.py)

These tests run directly against sleepwise.pl via pyswip, with no Python Bayesian engine involved. Each test asserts specific user_fact/2 clauses, queries the Prolog predicates, then checks the result. Total: 189 tests, all passed.

Section 1 — Cause Rules: Positive Cases (representative 39 of 81 tests)

Every cause in sleepwise.pl has at least one triggering clause. These tests verify each cause fires when its primary input fact is asserted. Both clauses of dual-clause causes (e.g. substance_use, irregular_schedule) are tested separately.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-001	high_caffeine fires	caffeine_intake=high	high_caffeine ∈ causes	high_caffeine ∈ causes	**PASS**
TC-KB-002	high_caffeine fires (medium)	caffeine_intake=medium	high_caffeine ∈ causes	high_caffeine ∈ causes	**PASS**
TC-KB-003	poor_diet_habits fires	heavy_meal_before_bed=often	poor_diet_habits ∈ causes	poor_diet_habits ∈ causes	**PASS**
TC-KB-005	sedentary_lifestyle fires	physical_activity_level=none	sedentary_lifestyle ∈ causes	sedentary_lifestyle ∈ causes	**PASS**
TC-KB-007	substance_use fires (alcohol)	alcohol_consumption=often	substance_use ∈ causes	substance_use ∈ causes	**PASS**
TC-KB-010	substance_use fires (drugs)	recreational_drug_use=frequently	substance_use ∈ causes	substance_use ∈ causes	**PASS**
TC-KB-011	poor_hydration fires	fluid_intake_timing=late	poor_hydration ∈ causes	poor_hydration ∈ causes	**PASS**
TC-KB-013	medication_interference fires	medication_supplements=daily_nonrx	medication_interference ∈ causes	medication_interference ∈ causes	**PASS**
TC-KB-015	high_stress fires	stress_level=high	high_stress ∈ causes	high_stress ∈ causes	**PASS**
TC-KB-016	anxiety fires	anxiety_levels=often	anxiety ∈ causes	anxiety ∈ causes	**PASS**
TC-KB-018	anxiety fires (panic)	panic_attacks=often	anxiety ∈ causes	anxiety ∈ causes	**PASS**
TC-KB-020	depression fires	depression_symptoms=often	depression ∈ causes	depression ∈ causes	**PASS**
TC-KB-022	burnout fires	burnout_feelings=often	burnout ∈ causes	burnout ∈ causes	**PASS**
TC-KB-024	rumination fires	rumination_habits=often	rumination ∈ causes	rumination ∈ causes	**PASS**

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-026	rumination fires (worry)	worry_before_bed=often	rumination ∈ causes	rumination ∈ causes	PASS
TC-KB-028	poor_self_control fires (sched)	consistent_sleep_time=inconsistent	poor_self_control ∈ causes	poor_self_control ∈ causes	PASS
TC-KB-029	poor_self_control fires (ADHD)	adhd_symptoms=moderate	poor_self_control ∈ causes	poor_self_control ∈ causes	PASS
TC-KB-031	perfectionism fires	perfectionism_tendencies=often	perfectionism ∈ causes	perfectionism ∈ causes	PASS
TC-KB-033	low_resilience fires	resilience_level=poorly	low_resilience ∈ causes	low_resilience ∈ causes	PASS
TC-KB-035	noise_disturbance fires	noisy_environment=loud	noise_disturbance ∈ causes	noise_disturbance ∈ causes	PASS
TC-KB-038	light_disturbance fires	light_exposure_at_night=often	light_disturbance ∈ causes	light_disturbance ∈ causes	PASS
TC-KB-042	temperature_discomfort fires	uncomfortable_temperature=too_hot	temperature_discomfort ∈ causes	temperature_discomfort ∈ causes	PASS
TC-KB-044	poor_sleep_space fires (clutter)	room_clutter=high	poor_sleep_space ∈ causes	poor_sleep_space ∈ causes	PASS
TC-KB-046	safety_concern fires	neighborhood_safety=moderate	safety_concern ∈ causes	safety_concern ∈ causes	PASS
TC-KB-048	excessive_screen_use fires	screen_before_bed=often	excessive_screen_use ∈ causes	excessive_screen_use ∈ causes	PASS
TC-KB-052	irregular_schedule fires	consistent_sleep_time=inconsistent	irregular_schedule ∈ causes	irregular_schedule ∈ causes	PASS
TC-KB-053	irregular_schedule fires (wknd)	weekend_sleep_catchup=much_more	irregular_schedule ∈ causes	irregular_schedule ∈ causes	PASS
TC-KB-055	poor_sleep_hygiene fires	bedtime_routine_consistency=rarely	poor_sleep_hygiene ∈ causes	poor_sleep_hygiene ∈ causes	PASS
TC-KB-057	excessive_napping fires	daytime_nap_hours=moderate	excessive_napping ∈ causes	excessive_napping ∈ causes	PASS
TC-KB-059	poor_sleep_efficiency (latency)	sleep_latency=slow	poor_sleep_efficiency ∈ causes	poor_sleep_efficiency ∈ causes	PASS
TC-KB-061	poor_sleep_efficiency (wakeups)	frequent_wakeups=several	poor_sleep_efficiency ∈ causes	poor_sleep_efficiency ∈ causes	PASS

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-063	poor_sleep_efficiency (disorder)	sleep_disorders=suspected	poor_sleep_efficiency ∈ causes	poor_sleep_efficiency ∈ causes	PASS
TC-KB-066	poor_sleep_duration fires	sleep_hours=less_than_4	poor_sleep_duration ∈ causes	poor_sleep_duration ∈ causes	PASS
TC-KB-068	academic_overload fires	academic_workload=heavy	academic_overload ∈ causes	academic_overload ∈ causes	PASS
TC-KB-071	late_night_studying fires	study_past_midnight=often	late_night_studying ∈ causes	late_night_studying ∈ causes	PASS
TC-KB-075	exam_pressure fires	exam_period=yes	exam_pressure ∈ causes	exam_pressure ∈ causes	PASS
TC-KB-078	performance_impact fires	performance_impact=often	performance_impact ∈ causes	performance_impact ∈ causes	PASS
TC-KB-080	performance_impact (attendance)	lecture_attendance=never	performance_impact ∈ causes	performance_impact ∈ causes	PASS

Section 2 — Cause Rules: Negative Cases (18 of 38 tests shown)

These tests verify that healthy or below-threshold values do not trigger any cause — including the fixed cases from Bug 3 (high_stress silent on stress_level=medium) and Bug 5 (performance_impact silent on lecture_attendance=often).

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-082	high_caffeine silent	caffeine_intake=none	high_caffeine ∉ causes	high_caffeine ∉ causes	PASS
TC-KB-083	high_caffeine silent (low)	caffeine_intake=low	high_caffeine ∉ causes	high_caffeine ∉ causes	PASS
TC-KB-084	sedentary silent (moderate)	physical_activity_level=moderate	sedentary_lifestyle ∉ causes	sedentary_lifestyle ∉ causes	PASS
TC-KB-086	substance_use silent	alcohol_consumption=never	substance_use ∉ causes	substance_use ∉ causes	PASS
TC-KB-090	high_stress silent (low)	stress_level=low	high_stress ∉ causes	high_stress ∉ causes	PASS
TC-KB-091	high_stress silent (medium)	stress_level=medium	high_stress ∉ causes	high_stress ∉ causes	PASS
TC-KB-092	anxiety silent	anxiety_levels=never	anxiety ∉ causes	anxiety ∉ causes	PASS
TC-KB-094	depression silent	depression_symptoms=never	depression ∉ causes	depression ∉ causes	PASS

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-096	burnout silent	burnout_feelings=never	burnout ∉ causes	burnout ∉ causes	PASS
TC-KB-098	poor_self_control silent (ADHD)	adhd_symptoms=no	poor_self_control ∉ causes	poor_self_control ∉ causes	PASS
TC-KB-099	poor_self_control silent (mild)	adhd_symptoms=mild	poor_self_control ∉ causes	poor_self_control ∉ causes	PASS
TC-KB-100	irregular/self-ctrl silent	consistent_sleep_time=very_consistent	both ∉ causes	both ∉ causes	PASS
TC-KB-101	irregular silent (same wknd)	weekend_sleep_catchup=same	irregular_schedule ∉ causes	irregular_schedule ∉ causes	PASS
TC-KB-107	sleep_efficiency silent (fast)	sleep_latency=fast	poor_sleep_efficiency ∉ causes	poor_sleep_efficiency ∉ causes	PASS
TC-KB-112	poor_sleep_duration silent	sleep_hours=six_to_eight	poor_sleep_duration ∉ causes	poor_sleep_duration ∉ causes	PASS
TC-KB-114	exam_pressure silent	exam_period=no	exam_pressure ∉ causes	exam_pressure ∉ causes	PASS
TC-KB-117	performance_impact silent	performance_impact=never	performance_impact ∉ causes	performance_impact ∉ causes	PASS
TC-KB-118	performance_impact silent (attend)	lecture_attendance=often	performance_impact ∉ causes	performance_impact ∉ causes	PASS

Section 3 — Severity Levels (6 tests)

Tests all four severity levels returned by sleep_severity/1: none, mild, moderate, severe. Severity is calculated by counting fired causes and checking anchor questions.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-120	Severity=none (no facts)	No facts asserted	none	none	PASS
TC-KB-121	Severity=mild (1 cause)	caffeine_intake=high	mild	mild	PASS
TC-KB-122	Severity=moderate (short sleep)	sleep_hours=four_to_six	moderate	moderate	PASS
TC-KB-123	Severity=severe (depression+sleep)	depression=every_day + sleep<4h	severe	severe	PASS
TC-KB-124	Severity=severe (sleep disorder)	sleep_disorders=diagnosed_severe	severe	severe	PASS
TC-KB-125	Severity=severe (4+ causes)	4 concurrent cause signals	severe	severe	PASS

Section 4 — Clinical Flags (20 tests)

All 5 clinical flags (CF1–CF5) tested for both positive and negative conditions. Each flag requires ALL conditions to be met — missing one condition must prevent the flag from firing.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-126	CF1 fires	depression=every_day + sleep<4h	CF1 ∈ flags	CF1 ∈ flags	**PASS**
TC-KB-128	CF1 silent (no short sleep)	depression=every_day + sleep=6-8h	CF1 ∉ flags	CF1 ∉ flags	**PASS**
TC-KB-129	CF1 silent (mild depr)	depression=rarely + sleep<4h	CF1 ∉ flags	CF1 ∉ flags	**PASS**
TC-KB-130	CF2 fires	anxiety=every_night + panic=often	CF2 ∈ flags	CF2 ∈ flags	**PASS**
TC-KB-132	CF2 silent (not every_night)	anxiety=often + panic=frequently	CF2 ∉ flags	CF2 ∉ flags	**PASS**
TC-KB-134	CF3 fires	sleep_disorders=diagnosed_severe	CF3 ∈ flags	CF3 ∈ flags	**PASS**
TC-KB-135	CF3 silent (mild)	sleep_disorders=diagnosed_mild	CF3 ∉ flags	CF3 ∉ flags	**PASS**
TC-KB-137	CF3 silent (none)	sleep_disorders=no	CF3 ∉ flags	CF3 ∉ flags	**PASS**
TC-KB-138	CF4 fires	burnout=constantly + depression=often	CF4 ∈ flags	CF4 ∈ flags	**PASS**
TC-KB-140	CF4 silent (no depr)	burnout=constantly + depression=never	CF4 ∉ flags	CF4 ∉ flags	**PASS**
TC-KB-141	CF4 silent (mild burnout)	burnout=often + depression=every_day	CF4 ∉ flags	CF4 ∉ flags	**PASS**
TC-KB-142	CF5 fires	depression + anxiety + sleep<4h	CF5 ∈ flags	CF5 ∈ flags	**PASS**
TC-KB-144	CF5 silent (sleep ok)	depression + anxiety + sleep=6-8h	CF5 ∉ flags	CF5 ∉ flags	**PASS**
TC-KB-145	CF5 silent (mild depr)	depression=rarely + anxiety + sleep<4h	CF5 ∉ flags	CF5 ∉ flags	**PASS**

Section 5 — Sleep Complaint Classification (7 tests)

Tests all 6 sleep complaint types returned by sleep_complaint/1, plus one negative case.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-146	sleep_deprivation	sleep_hours=less_than_4	sleep_deprivation ∈ complaints	sleep_deprivation ∈ complaints	**PASS**
TC-KB-148	insomnia	sleep_latency=very_slow	insomnia ∈ complaints	insomnia ∈ complaints	**PASS**
TC-KB-150	maintenance_insomnia	frequent_wakeups=several	maintenance_insomnia ∈ complaints	maintenance_insomnia ∈ complaints	**PASS**
TC-KB-152	hypersomnia	daytime_nap_hours=long + morning=extremely_tired	hypersomnia ∈ complaints	hypersomnia ∈ complaints	**PASS**
TC-KB-153	circadian_disruption	inconsistent schedule + screen use	circadian_disruption ∈ complaints	circadian_disruption ∈ complaints	**PASS**
TC-KB-154	stress_related_insomnia	stress_level=high + sleep_latency=very_slow	stress_related_insomnia ∈ complaints	stress_related_insomnia ∈ complaints	**PASS**

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-155	No complaint (healthy)	sleep_hours=six_to_eight	sleep_deprivation ∉ complaints	sleep_deprivation ∉ complaints	**PASS**

Section 6 — Recommendations (29 tests)

Every one of the 29 causes must have a recommendation/2 clause returning a substantive string.

Test IDs	Description	Input	Result
TC-KB-156–184	All 29 causes have recommendations	recommendation(Cause, _) for each of the 29 causes	**PASS**

Section 7 — Session Reset and Isolation (5 tests)

Verifies that reset_session/0 removes all asserted facts, that severity returns to none, and that facts from one session cannot bleed into the next.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-KB-185	Causes fire before reset	3 cause signals asserted	>0 causes	>0 causes	**PASS**
TC-KB-186	reset_session clears all facts	reset_session/0 called	user_fact count = 0	user_fact count = 0	**PASS**
TC-KB-187	No causes after reset	reset_session/0 called	causes = {}	causes = {}	**PASS**
TC-KB-188	Severity=none after reset	reset_session/0 called	severity = none	severity = none	**PASS**
TC-KB-189	No fact bleed between sessions	Session 1 facts retracted; new session	prev causes ∉ new session	prev causes ∉ new session	**PASS**

PART 2 — System Integration Tests (test_system_integration.py)

These tests exercise the complete pipeline: Python (question_bank → bayes_engine → session_manager) connected to Prolog (sleepwise.pl) via the pyswip bridge. Total: 78 tests, all passed.

IT-01 — Prolog Bridge Initialisation (5 tests)

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-001	pyswip library importable	import pyswip	No ImportError	Imported successfully	**PASS**
TC-IT-002	sleepwise.pl found at expected path	../knowledge_base/sleepwise.pl	File exists	File exists	**PASS**
TC-IT-003	Prolog instance initialises	Prolog(); consult(sleepwise.pl)	No exception	Instance created	**PASS**
TC-IT-004	Basic query (true) succeeds	list(p.query('true'))	Returns result	Returns [{}]	**PASS**
TC-IT-005	retractall/1 executes	retractall(user_fact(_, _))	No exception	Executes cleanly	**PASS**

IT-02 — Python assert → Prolog Query Round-trip (5 tests)

Facts asserted from Python via assertz must be immediately queryable from Prolog. Verifies the assert/retract cycle that the main session loop depends on.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-006	Asserted fact is queryable	assertz(user_fact(caffeine_intake,high))	Queryable via user_fact	Query returns 1 result	**PASS**
TC-IT-007	All 4 facts retrievable	4 facts asserted, query all	4 facts stored	4 facts retrieved	**PASS**
TC-IT-008	retractall clears all facts	retractall after 4 facts	Count = 0	Count = 0	**PASS**
TC-IT-009	Atom values assert cleanly	assertz(user_fact(academic_workload,heavy))	No syntax error	Asserted successfully	**PASS**
TC-IT-010	10-fact assertion = 10 stored	10 facts asserted via loop	Count = 10	Count = 10	**PASS**

IT-03 — Python Engine and Prolog Cause Agreement (12 tests)

For each of 10 causes, both the Python Bayesian engine and Prolog must identify the cause when given inputs strong enough to trigger both. Note: Python uses a distribution-relative threshold requiring multi-signal corroboration; Prolog fires on a single matching fact. Test inputs therefore use all likelihood-relevant questions at triggering values.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-011	Both identify depression	depression=every_day + loneliness=constantly + trauma=significant + sleep<4h + morning=extremely_tired + optimism=pessimistic	depression in both	depression in both	**PASS**
TC-IT-012	Both identify high_stress	stress_level=high + daytime_fatigue=every_day	high_stress in both	high_stress in both	**PASS**

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-013	Both identify substance_use	alcohol=daily + drugs=often	substance_use in both	substance_use in both	PASS
TC-IT-014	Both identify anxiety	anxiety=every_night + worry=always + panic=often	anxiety in both	anxiety in both	PASS
TC-IT-015	Both identify burnout	burnout=constantly + motivation=very_low + fatigue=every_day + morning=extremely_tired	burnout in both	burnout in both	PASS
TC-IT-016	Both identify academic_overload	workload=heavy + study_midnight=often + exam=yes	academic_overload in both	academic_overload in both	PASS
TC-IT-017	Both identify noise_disturbance	noisy=loud + snoring=often	noise_disturbance in both	noise_disturbance in both	PASS
TC-IT-018	Both identify excessive_screen_use	screen=always + social_media=long	excessive_screen_use in both	excessive_screen_use in both	PASS
TC-IT-019	Both identify late_night_studying	study_midnight=always + exam_prep=always	late_night_studying in both	late_night_studying in both	PASS
TC-IT-020	Both identify irregular_schedule	consistent_sleep=inconsistent + wknd=far_more	irregular_schedule in both	irregular_schedule in both	PASS
TC-IT-021	Healthy: Prolog fires 0 causes	All 120 questions at healthy defaults	causes = {}	causes = {}	PASS
TC-IT-022	Healthy: Python fires 0 causes	All 120 questions at healthy defaults	fired_causes = {}	fired_causes = {}	PASS

IT-04 — Severity Agreement (9 tests)

Prolog and Python both tested for severity accuracy across key scenarios. sleep_disorders=diagnosed_severe is only included in q50 cores and is therefore tested with the q50 variant only.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-023	Prolog: none (empty session)	No facts asserted	none	none	PASS
TC-IT-024	Python: none (healthy defaults)	All 120 healthy values	none	none	PASS
TC-IT-025	Prolog: severe (depr+sleep)	depression=every_day + sleep=less_than_4	severe	severe	PASS
TC-IT-026	Python: severe (depr+sleep)	depression=every_day + sleep=less_than_4	severe	severe	PASS
TC-IT-027	Prolog: severe (disorder, q50)	sleep_disorders=diagnosed_severe (q50)	severe	severe	PASS
TC-IT-028	Prolog: severe (multi-cause)	stress=high + sleep<4h + depr + anxiety	severe	severe	PASS
TC-IT-029	Python: severe (multi-cause)	stress=high + sleep<4h + depr + anxiety	severe	severe	PASS

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-030	Prolog: mild (1 cause)	caffeine_intake=high only	mild	mild	**PASS**
TC-IT-031	Prolog: none (no facts)	No facts asserted	none	none	**PASS**

IT-05 — Recommendation Retrieval (12 tests)

Recommendations must be retrievable through the bridge for all 29 causes. Spot-checks for 10 causes, full check for all 29, and graceful handling of unknown causes.

Test IDs	Description	Input	Expected	Actual	Result
TC-IT-032–041	10 spot-check recommendations	recommendation(C,_) for 10 representative causes	Non-empty string >20 chars	All 10 returned non-empty strings	**PASS**
TC-IT-042	All 29 recs substantive (>50 chars)	recommendation(C,_) for all 29 causes	Length ≥50 for all	All 29 passed length check	**PASS**
TC-IT-043	Unknown cause returns empty	recommendation(nonexistent_xyz, R)	Empty string	Empty string	**PASS**

IT-06 — Full Session Pipeline, All 4 Variants (20 tests)

End-to-end: a stressed student persona passes through each of the four session variants (q20/q30/q40/q50). Checks question count ceiling, Python cause detection, Prolog cause detection, severity, and recommendation availability. Clinical add-ons are outside the standard question budget by design — ceiling = total_questions + 9.

Test IDs	Variant	Input Persona	All Checks	Result
TC-IT-044–048	[q20] Full pipeline	Stressed persona through q20 variant	≤24 Qs; ≥1 cause; severity≠none; recs exist	**PASS**
TC-IT-049–053	[q30] Full pipeline	Stressed persona through q30 variant	≤34 Qs; ≥1 cause; severity≠none; recs exist	**PASS**
TC-IT-054–058	[q40] Full pipeline	Stressed persona through q40 variant	≤44 Qs; ≥1 cause; severity≠none; recs exist	**PASS**
TC-IT-059–063	[q50] Full pipeline	Stressed persona through q50 variant	≤59 Qs; ≥1 cause; severity≠none; recs exist	**PASS**

IT-07 — Multi-Session Isolation (6 tests)

Session N facts must not affect session N+1. Also verifies the Python engine is deterministic — running the same session twice returns identical results.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-064	Session 1 severity=severe	depression=every_day + sleep=less_than_4	severity=severe	severity=severe	**PASS**
TC-IT-065	Session 1 depression identified	Same as above	depression ∈ causes	depression ∈ causes	**PASS**
TC-IT-066	Session 2: no bleed (severity)	retractall; healthy facts	severity=none	severity=none	**PASS**
TC-IT-067	Session 2: no bleed (depression)	retractall; healthy facts	depression ∉ causes	depression ∉ causes	**PASS**
TC-IT-068	Repeated run: identical severity	Same answers, run_session ×2	same severity	same severity	**PASS**

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-069	Repeated run: identical causes	Same answers, run_session ×2	same fired_causes	same fired_causes	**PASS**

IT-08 — Graceful Degradation, Python-only Mode (9 tests)

The system must complete a full diagnosis and return a valid result structure even when Prolog is unavailable. All four variants tested.

Test ID	Description	Input / Condition	Expected	Actual	Result
TC-IT-070	Diagnosis keys present	run_session() without Prolog	severity, fired_causes, complaints, severity_score ∈ result	All 4 keys present	**PASS**
TC-IT-071	Valid severity value	run_session() result	Value ∈ {none,mild,moderate,severe}	Valid value returned	**PASS**
TC-IT-072	fired_causes is dict	type(diag['fired_causes'])	dict	dict	**PASS**
TC-IT-073	complaints is list	type(diag['complaints'])	list	list	**PASS**
TC-IT-074	severity_score in [0,1]	0.0 ≤ severity_score ≤ 1.0	[0.0, 1.0]	In range	**PASS**
TC-IT-075–078	All 4 variants complete	run_session(variant) for q20/30/40/50	No exception; 'diagnosis' in result	All 4 completed	**PASS**

Test Execution Summary

Script	Sections	Total Tests	Passed	Failed	Result
test_knowledge_base.py	7	189	189	0	**PASS**
test_system_integration.py	8	78	78	0	**PASS**
TOTAL	**15**	**267**	**267**	**0**	**PASS**

All 267 test cases passed. The knowledge base, Bayesian engine, and Python-Prolog integration have been verified end-to-end across all four session variants.

Test scripts: interface/test_knowledge_base.py | interface/test_system_integration.py
Knowledge base: knowledge_base/sleepwise.pl | DCIT 313 Group KENS